

Deepak Mallampati

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Kent, OH - F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in LLM-powered automation pipelines, RAG architectures, and API-driven workflow orchestration. Built production multi-agent systems with conditional routing, error handling, webhook-triggered processing, and structured JSON contracts between stages - the same orchestration patterns that n8n abstracts at the visual layer. Strong background in Python, TypeScript, REST APIs, and LLM prompt engineering. Experienced with RAG data integration and structured output design. F-1 OPT; open to Germany-based sponsorship.

CORE COMPETENCIES

LLM Workflow Automation • Retrieval-Augmented Generation (RAG) • API & Webhook Orchestration • Structured Outputs & Prompt Engineering • Python / TypeScript / JavaScript • Multi-Agent Pipeline Design • Error Handling & Observability • FastAPI / Docker • Data Integration Patterns • LangChain / LlamaIndex • Healthcare AI (HIPAA) • CI/CD

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS

- Built **production RAG systems** for clinical knowledge retrieval with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, retrieval-grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning and tool discipline; improved agent response stability **~25%** and reduced hallucinated outputs **>30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall **15-20%** on high-risk categories at **>90%** precision.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services with Docker, structured logging, and load simulation; reduced post-deployment defects **~30%**.
- Maintained HIPAA-conscious data governance with de-identification, data lineage, and evaluation audit trails.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures for Order-to-Cash platform; reduced query latency **~25%** and execution time **~20%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and SQL deployments; reduced deployment errors **>30%**, improved release cycle **~35-40%**.
- Designed Grafana dashboards tracking long-running queries, deadlocks, and I/O contention; cut incident recurrence **~25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics

- Built transformer-based video summarization system across 5,000+ sessions; **60-70% reduction** in manual review time, **~85% alignment** with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs **~30%**, raised contextual accuracy **>85%**.

Kent State University

Aug 2023 - May 2025

Student Manager - Operations, Team Leadership

- Led coordination for **150+ students**; reduced coordination delays **~30%** and formalized onboarding cutting ramp-up **~20%**.

FEATURED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready traces.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, agentic orchestration

Dream Decoder - Multimodal AI Creative Platform

FastAPI + React/TypeScript/Vite/Tailwind app coordinating multiple AI APIs (Perplexity Sonar, Replicate image models). Intermediate structured prompt transformation layers improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%** over naive direct-prompt approaches.

FastAPI, React, TypeScript, multi-LLM orchestration, structured outputs, REST APIs

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Full-stack Manifest V3 extension integrating four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider webhook-style routing with per-domain selector configs for 29 sites, seven-day translation cache, and viewport-slice screenshot pipeline with coordinate remapping.

TypeScript, React, Manifest V3, multi-provider API orchestration, WebP/JPEG routing

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN with unified YOLOv8; reduced inference latency ~**30%**. Sliding-window confidence aggregation suppressed blink-driven false positives ~**25%**; adaptive frame skipping and NMS tuning for stable real-time OpenCV capture.

Python, YOLOv8, OpenCV, PyTorch, LabelImg, real-time inference

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL), FastAPI, Flask

LLM & Automation: LLM APIs, RAG, agentic workflows, structured outputs, webhook integrations, conditional routing, multi-step pipeline design

AI Libraries: LangChain, LlamaIndex, Hugging Face Transformers, scikit-learn, PyTorch

Data & APIs: REST APIs, JSON schema validation, Pandas, data preprocessing, EHR extract pipelines

Infrastructure: Docker, Jenkins CI/CD, Grafana, FastAPI services, cloud-ready deployment